

Assignment 2 – Car Price Prediction

Problem Statement

The automotive industry is highly dynamic, with car prices influenced by multiple factors, including engine type, fuel system, and body style. This project will utilize **exploratory data analysis (EDA)** and **machine learning modeling** to investigate the factors affecting car prices. The goal is to derive insights into price variations and develop a predictive model for car pricing.

Data

The dataset for this project includes various attributes related to cars, such as engine specifications, fuel type, and structural features. Key variables include:

- Price
 - Cylinder number
 - Number of doors
 - Engine type
 - Fuel system
 - Aspiration
 - Car body
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Instructions

Load the dataset and complete the following tasks:

Exploratory Data Analysis (EDA)

- 1. Price Variation by Cylinder Number (Gas Cars)**
 - Analyze how the price of gas-powered cars varies with the number of cylinders.
 - Provide a **graph** illustrating this variation and a detailed explanation of insights.
- 2. Impact of Number of Doors on Price (Gas Cars)**
 - Investigate whether the number of doors influences the price of gas-powered cars.
 - Present findings with appropriate visualizations and interpretation.
- 3. Effect of Engine Type on Price (Gas Cars)**
 - Analyze whether different engine types impact pricing.
 - Provide insights with supporting visualizations.

4. Influence of Fuel System, Aspiration, and Car Body (Gas Cars)

- Examine how fuel system type, aspiration, and car body design impact price.
- Use graphs and charts to justify conclusions.

5. Repeat the Above Analysis for Diesel Cars

- Conduct the same analysis as in **points 1–4**, but for **diesel-powered cars**.
- Compare key differences between gas and diesel vehicles.

6. Feature Statistics

- Compute **summary statistics** (mean, median, standard deviation, min, max) for numerical features.
- Provide detailed observations in **plain English**.

7. Correlation Analysis

- Identify the correlation between different car attributes.
- Highlight key insights and dependencies among variables.

8. Violin and Box Plot Analysis

- Use **violin plots** and **box plots** to analyze distributions of numerical features.
- Summarize findings in a structured format.

9. Correlation Matrix & Widget Analysis

- Generate a **correlation matrix** and use correlation widgets to observe relationships.
- Present critical observations based on findings.

Machine Learning Modeling

1. Linear Regression Model – Base Approach

- Use a **data sampler widget** to **split** the dataset (70% train, 30% test).
- Train a **linear regression model** and evaluate performance.
- Change the **cross-validation type** (random sampling, leave-one-out) and observe changes in R^2 .
- **Document results in a structured table.**
- Extract **model coefficients** and explain their influence on price.
- Use the **prediction widget** to generate **predicted values**.

2. Feature Selection & Model Improvement

- Use a **feature ranking widget** to select the most influential features.
- Train a **linear regression model** with the selected features and compare performance.
- Change the **cross-validation type** (random sampling, leave-one-out) and analyze changes in R^2 .
- **Present results in a structured table.**
- Extract and explain the **model coefficients** and their impact on price.
- Use the **prediction widget** to generate and report predicted values.

Submission Guidelines

Complete the **presentation file** with the following format and email to **[Professor's Email]**:

1. **Slide 1** – Business Problem
2. **Slide 2** – Data Dictionary
3. **Slides 3–9** – Analysis for all 9 questions (Graphs, Charts, and Explanations)
4. **Slide 10** – Screenshot of **Orange Workflow**
5. **Slide 11** – Final Observations

Additionally, submit the **.ows (Orange workflow) file** alongside your PowerPoint.

File Naming Convention:

- **StudentName_Assignment2.ppt**
- **StudentName_Assignment2.ows**